

Physiology: Transport of Potassium, Calcium, Magnesium & Phosphate

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Session Objectives

At the conclusion of the session, the student should be able:

- 1. Define & describe the factors influencing K^+ balance.**
- 2. Define & describe the renin-angiotensin-aldosterone system and its role in K^+ balance in the nephron.**
- 3. Interpret the relationships between acid/base balance and K^+ homeostasis.**
- 4. Describe the regulation of K^+ , Ca^{++} , Mg^{++} & PO_4^{3-} in the nephron.**

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You've just spent your last "free" summer trying to get back into shape, as your diet and exercise habits in your 1st year of medical school were horrible. It has been a very hot summer with multiple heat waves.

One very hot afternoon (Heat Index of 103.5°), you decide to go jogging after a session practicing hot yoga. When you leave the studio, you are light-headed and you feel your heart "skipping beats".

Your yoga instructors give you a couple of bananas and a bottle of coconut water. After about 5', you feel better. You wonder what was in the bananas and coconut water that magically helped you recover so quickly.



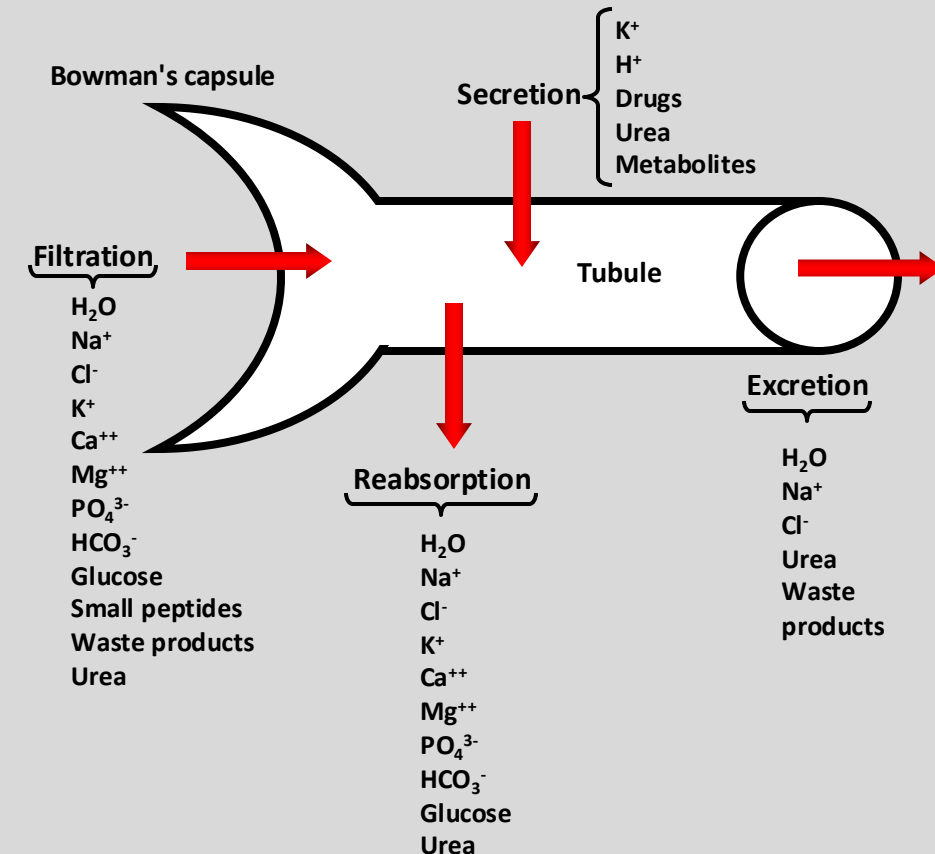


The Nephron

- The kidney plays a central role in controlling the plasma levels of a wide range of solutes
- All are present at low concentrations in the body
- The renal excretion of a solute depends on three processes: Filtration, Reabsorption & Secretion
- The kidney filters and then reabsorbs substances, some totally (e.g., glucose).
- It also filter & secrete (e.g., K^+ , organic anions & cations)
- It can also filter, reabsorb, and secrete (e.g., urea).
- The rates at which different substances are excreted represent the sum of these three renal processes

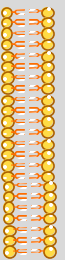
Urinary excretion rate = Filtration rate – Reabsorption rate + Secretion rate

- In general, tubular reabsorption is quantitatively more important than tubular secretion in urine formation
- Secretion plays an important role in determining the amounts of K^+ , H^+ & other substances.
- Electrolytes (e.g. Na^+ , Cl^- , HCO_3^-) are highly reabsorbed.



K⁺ Balance & Renal Regulation of K⁺

- The distribution of K⁺ differs from that of Na⁺
- ~98% of the total-body K⁺ is ICF vs 2% in the extracellular fluid (ECF).
- Plasma [K⁺] is tightly maintained at 3.5-5.0 mM.
- Chronic hypokalemia leads to several metabolic disturbances:
 - Inability of the kidney to form concentrated urine
 - Metabolic alkalosis
 - Enhancement of renal excretion of NH₃
- Elevated [K⁺]_{IC} is important for:
 - Maintenance of cell volume
 - Enzyme function
 - DNA and protein synthesis
 - Cell growth
- Low [K⁺]_{EC}:
 - Integral for membrane potential of excitable and non-excitable cells.
- Changes in [K⁺] ratios can cause disturbances in excitation and contraction of muscles.
- Doubling the plasma [K⁺]_{PI}/Reduction by half:
 - Severe disturbances in skeletal and cardiac muscle function.
 - Potentially life-threatening disturbances of cardiac rhythmicity from hyperkalemia

ECF/Plasma	Plasma Membrane	Intracellular Fluid
$[Na^+] \cong 142 \text{ mM}$ $[K^+] \cong 4.4 \text{ mM}$ $[Ca^{++}] \cong 2.4 \text{ mM}$ $[Mg^{++}] \cong 1 \text{ mM}$ $[PO_3^-] \cong 1 \text{ mM}$ $[Cl^-] \cong 102 \text{ mM}$		$[Na^+] \cong 15 \text{ mM}$ $[K^+] \cong 125 \text{ mM}$ $[Cl^-] \cong 20 \text{ mM}$ $[Mg^{++}] \cong 20 \text{ nM}$ $[PO_3^-] \cong 5-70 \text{ mM}$ $[Ca^{++}] \cong 100 \text{ nM}$

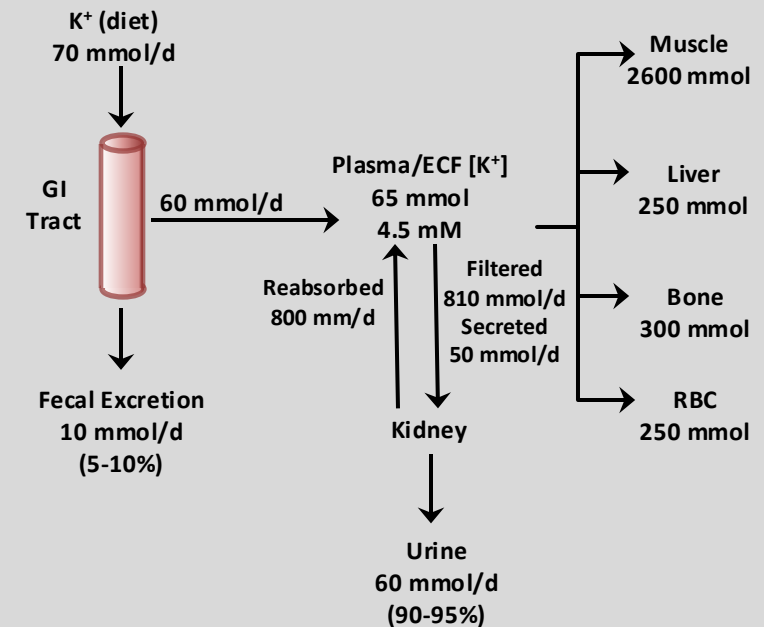
Intracellular [K ⁺] ([K ⁺] _{IC})		
	↓ [K ⁺] _{IC}	↑ [K ⁺] _{IC}
Cell volume	Shrinkage	Swelling
DNA/protein synthesis, growth	Reduction	+/-

Extracellular [K ⁺] ([K ⁺] _{EC})		
	↓ [K ⁺] _{EC}	↑ [K ⁺] _{EC}
pH _{IC} regulation	Acidosis	Alkalosis
Vascular tone	Constriction	Relaxation
Neuromuscular Activity	Muscle weakness & paralysis, resp. failure, abdominal distension	Increase excitability, eventual flaccid paralysis
Cardiac activity	prolonged repolarization, slowed conduction, abnormal pacemaker activity, tachyarrhythmia	enhanced repolarization, slowed conduction, bradyarrhythmia,

Intracellular [K ⁺]/Extracellular [K ⁺] ([K ⁺] _{IC} /[K ⁺] _{EC})		
	↓ ([K ⁺] _{IC} /[K ⁺] _{EC})	↑ ([K ⁺] _{IC} /[K ⁺] _{EC})
Resting membrane Pot.	Depolarization	Hyperpolarization

K⁺ Homeostasis

K⁺ Distribution & Balance



- Homeostasis involves external K⁺ (environment vs. body) & internal K⁺ balance (IC vs EC)
- External K⁺ Balance:
 - Determined by the relationship between [K⁺] intake & excretion.
 - [K⁺] intake = [K⁺]_{EC} (~ 65-75 mmol)
 - For [K⁺]_{EC} to remain constant, the body must excrete [K⁺] via renal & extrarenal mechanisms at the same rate as ingestion.
 - [K⁺] excretory mechanisms must be able to adjust to variable intake.
 - The kidney is largely responsible for [K⁺] excretion
 - GI tract plays a minor role.
 - GI tract can adjust [K⁺] excretion in response to:
 - Adrenal hormones
 - Δ in [K⁺] intake
 - Decreased capacity of the kidneys to excrete [K⁺].
- Internal K⁺ Balance:
 - Majority of total body [K⁺] is intracellular.
 - Unequal distribution between IC & EC [K⁺] has important implications.
 - Total [K⁺] ~ 3000 mmol
 - Shuttling as little as 1% to or from the ECF would cause a 50% change in extracellular [K⁺]
 - Severe consequences for neuromuscular function

K⁺ Homeostasis

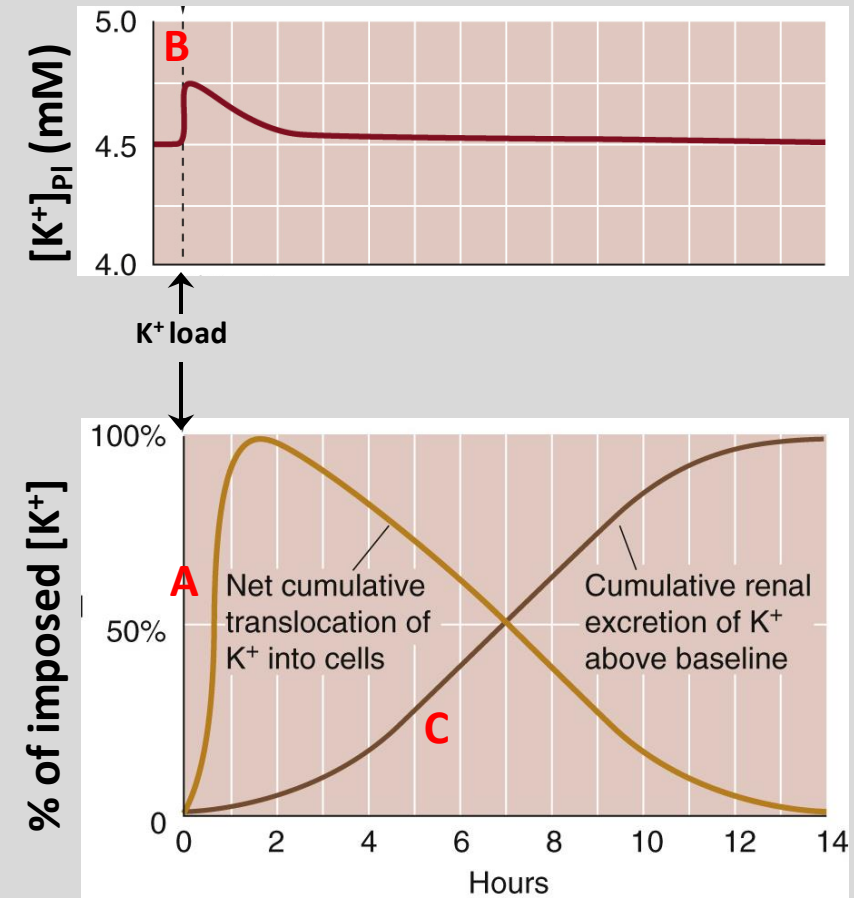
- Most common source of daily intake is dietary (via intestinal absorption)
- Substantial amounts of K⁺ may also enter the ECF from damaged tissues.

After K⁺ ingestion:

- ~ 80% of ingested K⁺ load temporarily moves into cells **(A)**.
- Modest increase in [K⁺]_{pl} **(B)**.
- Transfer of excess K⁺ into cells: Rapid **(A)**.
- After a delay, the kidneys begin to excrete [K⁺]_{pl} **(C)**.
- The small increases in [K⁺]_{pl} → Actions of insulin, epinephrine (Epi), and aldosterone (Aldo) on target tissues.
- Promote the transfer of [K⁺]_{pl} to [K⁺]_{ic} via the Na⁺/K⁺ ATPase.
- Insulin:
 - ↑ [K⁺]_{pl} → Depolarization of pancreatic β cells → insulin release.
- Epinephrine:
 - Pathophysiological increase in [K⁺]_{pl} → ↑ Epi secretion
 - Physiological increase in [K⁺]_{pl} → No Δ in Epi secretion.
- Aldosterone:
 - ↑ [K⁺]_{pl} → Depolarization of adrenal glomerulosa cells → ↑ Aldo release

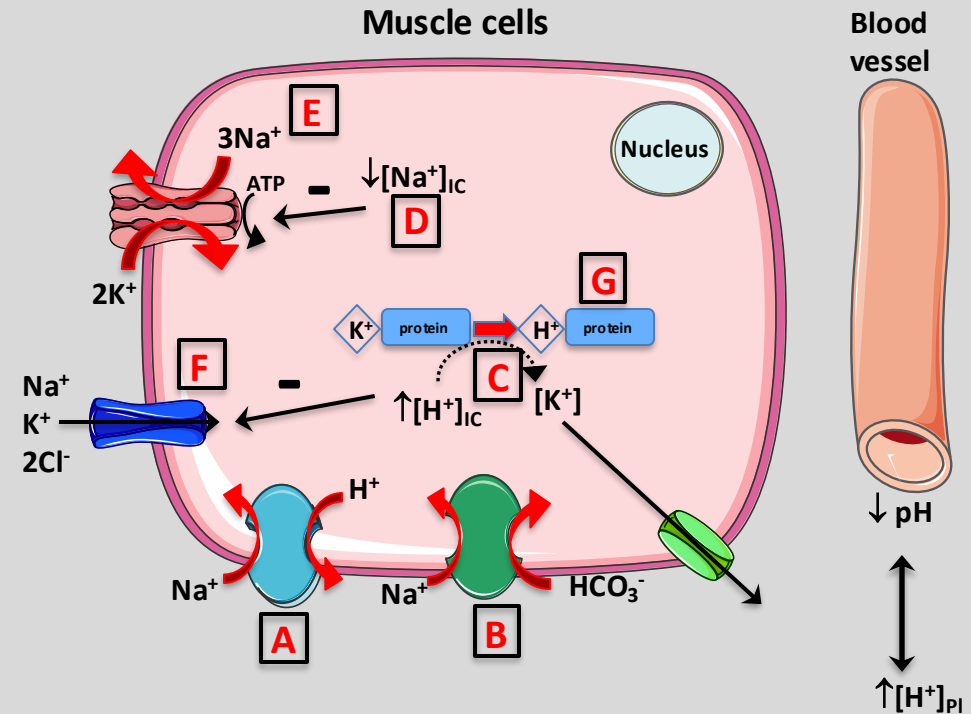
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K⁺ Homeostasis-Acidosis

- Acid-base disturbances also affect internal K⁺ distribution.
- Acidemia:
 - $\uparrow [H^+]_{EC}$ ($\downarrow pH_{EC}$) inhibits Na⁺/H⁺ exchanger and Na⁺/HCO₃⁻ cotransporter (**A&B**)
 - Results in $\uparrow [H^+]_{IC}$ (**C**) & $\downarrow [Na^+]_{IC}$ (**D**)
 - The intracellular acidosis compromises both the Na⁺-K⁺ ATPase (**E**) and the Na⁺/K⁺/2Cl⁻ cotransporter (NKCC) (**F**).
 - This will inhibit K⁺ influx.
 - $\downarrow pH_{IC}$ lessens the binding of K⁺ to nondiffusible intracellular anions (**G**).
 - This will promote K⁺ efflux.
 - These mechanisms all promote HYPERKALEMIA.



- Acidosis: PROCESS that causes an abnormal buildup of acid in the body
- Acidemia: STATE of abnormally high acidity in the blood ($pH < 7.35$)

K⁺ Homeostasis-Alkalosis

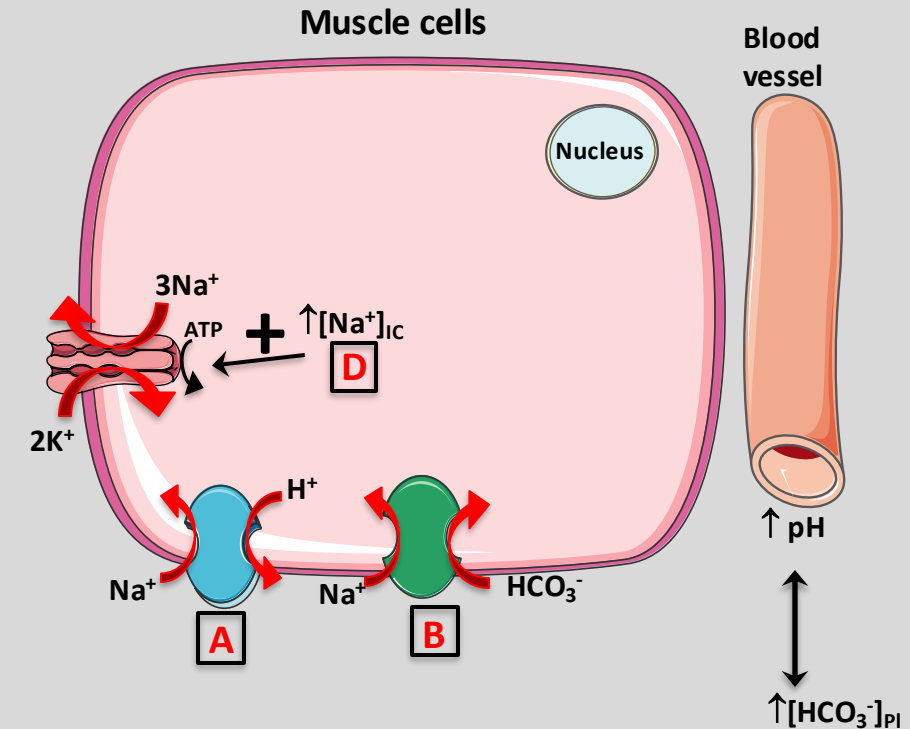
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- Alkalemia: Causes cells to take up K⁺ → HYPOKALEMIA.
- ↑pH and ↑[HCO₃⁻] enhance Na⁺ entry into the cell via Na⁺-H⁺ exchanger (**A**) and Na⁺/HCO₃⁻ cotransporter (**B**).
- ↑[Na⁺]_{IC} (**D**) → Stimulation of the Na⁺-K⁺ATPase → ↑K⁺ influx
- The opposite side of the coin, appearing as the exchange of K⁺ for H⁺

The effect of changes in extracellular [K⁺]_{EC} on acid-base homeostasis.

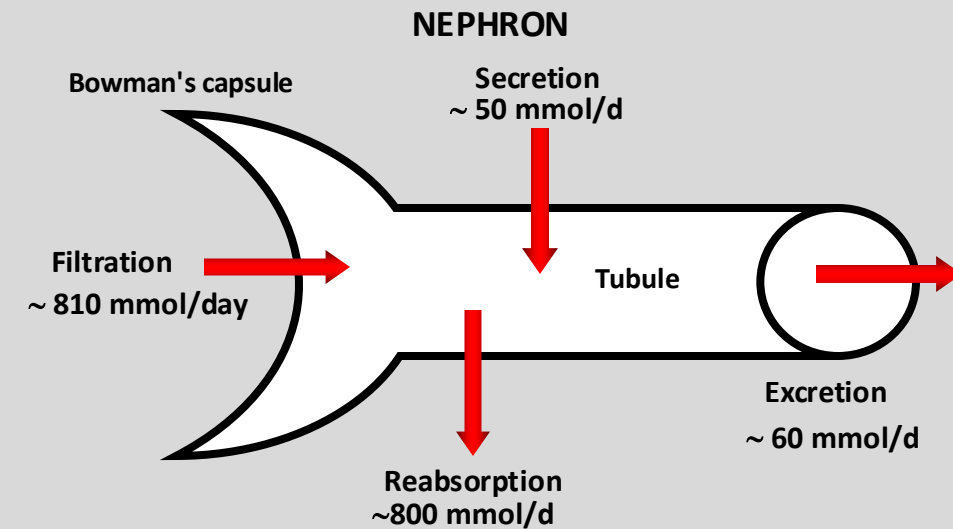
- Hyperkalemia causes intracellular alkalosis and extracellular acidosis.
- Hypokalemia causes intracellular acidosis and extracellular alkalosis.



- Alkalosis: PROCESS that causes increases in blood pH
- Alkalemia: STATE of abnormally high blood pH (pH > 7.45)

Total K⁺ Homeostasis

- The kidney excretes K⁺ via a combination of filtration, reabsorption, and secretion.
- In a healthy person (normal GFR and [K⁺]_{pl}):
 - Normal K⁺ dietary intake: 60-80 mmol/day.
 - Kidney filters ~ 800 mmol/day
- To achieve K⁺ balance, the kidneys need to excrete 10-15% of the filtered K⁺ (Filtered Load, FL).
- Low dietary K + intake:
 - 1-3% of the FL of K⁺
- With a normal or low K⁺ diet:
 - Kidneys can achieve K⁺ balance by filtration and reabsorption alone.
- Chronic high K⁺ intake:
 - Kidneys must rid the body of excess K⁺
 - K⁺ urinary excretion may exceed 150% of the filtered load.
- In the absence of a large dietary FL:
 - K⁺ tubular secretion is an important component of urinary excretion.
- K⁺ handling is a complex combination of glomerular filtration as well as reabsorption and secretion by the tubule.



K⁺ Transport in the Nephron

Pattern of K⁺ transport along the nephron under low or normal/high K⁺.

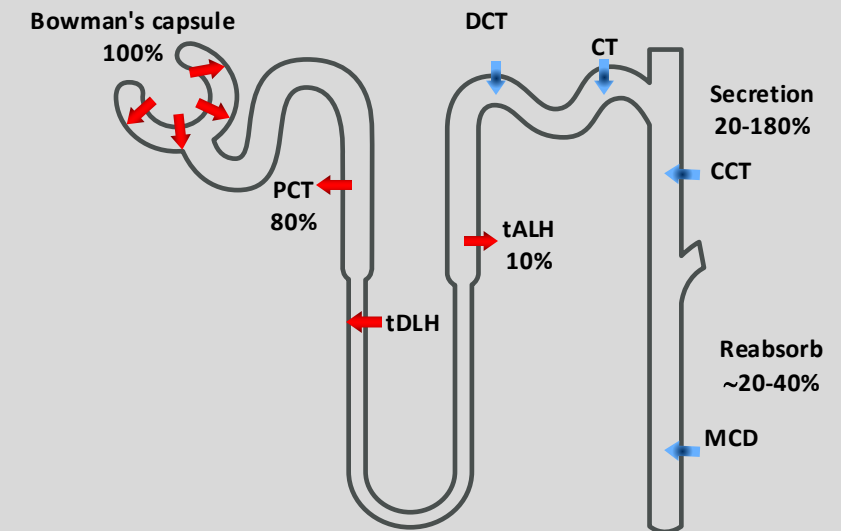
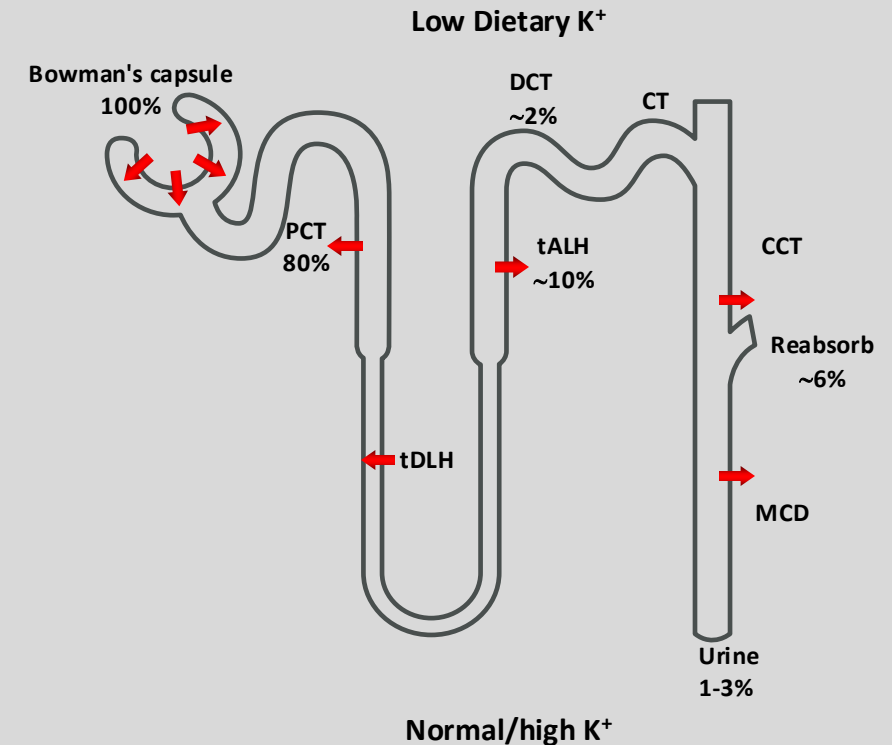
- In both cases, the kidney filters K⁺ in the glomerulus.
- ~80% is reabsorbed (isosmotically) in the PCT
- ~10% is reabsorbed tALH.
- ~10% of the FL enters the DCT.
- Urinary K⁺ content depends on the distal K⁺ secretory system :
 - DCT
 - Connecting tubule (CT)
 - Cortical collecting tubule (CCT), & medullary collecting duct (MCD).

Low Dietary K⁺:

- Body is trying to conserve K⁺
- DCT, CNT, ICT & CCT all reabsorb K⁺
- Only a small fraction of FL (1% to 3%) appears in the urine
- Not as effective as Na⁺ reabsorption.

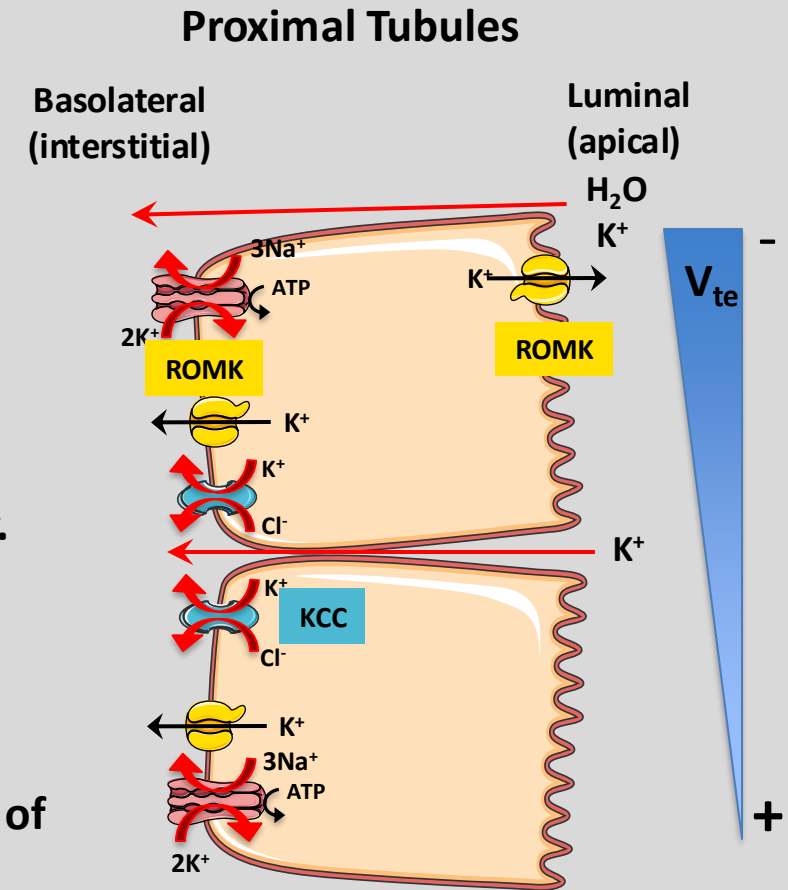
Normal or High Dietary K⁺:

- ICT, CCT, and the more proximal portion of the MCD secrete K⁺.
- Account for most of the urinary excretion of K⁺.
- This system responds to many stimuli that modulate K⁺.
- Any urinary K⁺ is largely represented by K⁺ secreted by more distal segments of the nephron.



K⁺ Reabsorption in the Proximal Tubule

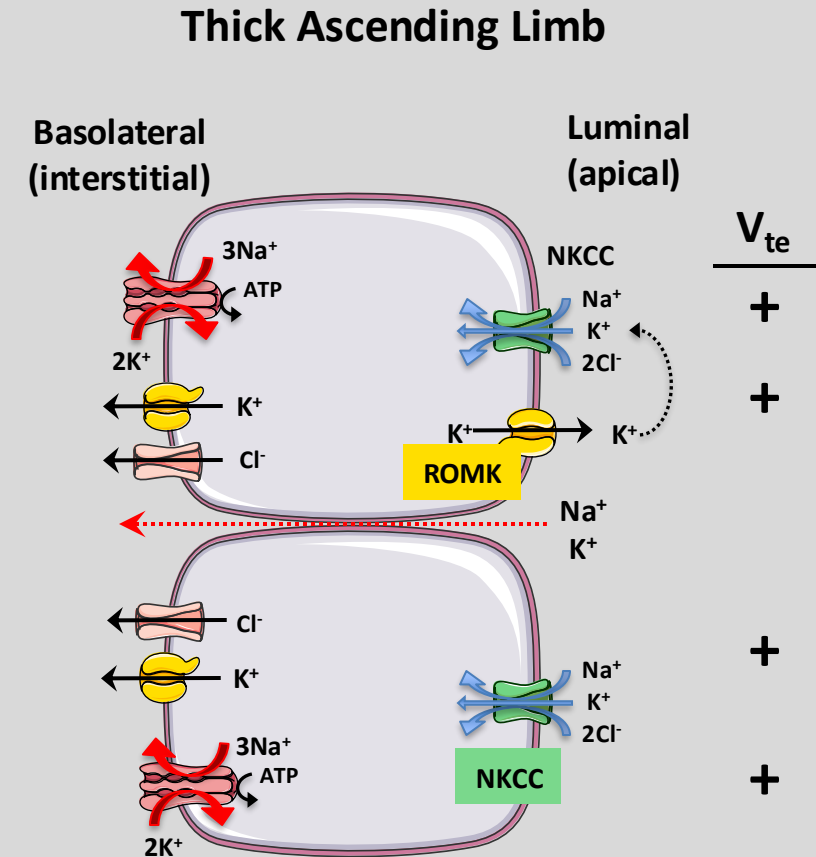
- The PCT reabsorbs most of the filtered K⁺
- This process occurs via two paracellular mechanisms:
 - **Electrodiffusion:**
 - Dependent on the luminal [K⁺] & transepithelial voltage (V_{te}).
 - H₂O reabsorption increases luminal [K⁺]
 - This generates a transepithelial gradient that drives passive absorption
 - As fluid flows down the PCT tubule, V_{te} shifts from negative to positive
 - This favors K⁺] reabsorption by electrodiffusion via paracellular pathway.
 - **Solvent drag:**
 - Diffusive movement of solutes is the dominant mode of transcapillary exchange.
 - The convective movement of H₂O can also carry solutes.
 - The flux of a dissolved solute that is swept along by the bulk movement of the solvent.
- K⁺ conductance of the basolateral membrane >>> apical membrane
- Basolateral K⁺ channels: Inwardly-rectifying K⁺ channels (renal outer medullary potassium channel (ROMK) channels)
 - Characterized by ATP sensitivity
 - ↓ [ATP] relieves inhibition of the K⁺ channels
- Apical (luminal) K⁺ channels are quiescent under normal conditions



Nephron Segment	Transepithelial Voltage (mV)
PCT (S1)	-3
PCT (S3)	+3
TAL	+6
DCT	-24 to -14
CCT	-74

K⁺ Reabsorption in the Thick Ascending Limb

- The descending limb of the loops of Henle (tDLH) *secrete* K⁺.
- Secretion is passive, driven by the high [K⁺] of the medullary interstitium & elevated [K⁺] permeability.
- K⁺ Reabsorption in the tALH:
 - Paracellular pathway:
 - Passive reabsorption
 - Driving force is the lumen-to-interstitium concentration gradient.
 - Minor pathway.
 - Transcellular pathway:
 - Majority of K⁺ use the apical Na⁺/K⁺/2Cl⁻ co-transporter.
 - Removing any of the ions from the lumen abolishes reabsorption.
 - Inhibition of NKCC abolishes the lumen-positive V_{te}.
 - This will also block paracellular K⁺ reabsorption.
- NKCC is sensitive to high-ceiling class of diuretics (Loop Diuretics).
- Apical K⁺ conductance occurs via ROMK channels.
- Major function of ROMK channel:
 - Mechanism for K⁺ recycling

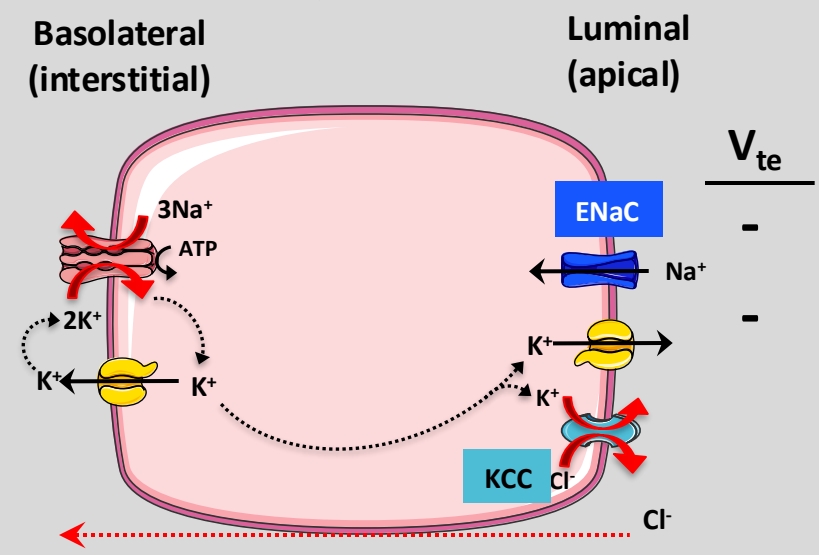


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K⁺ Reabsorption in the Collecting Tubule

- Urinary K⁺ content depends on the distal K⁺ secretory system (DCT, CT, CCT & MCD).
- Ultrastructurally, the ICT and CCT are nearly identical
- They are made up of:
 - Principal cells (~70%):
 - Secrete K⁺ into the tubular fluid
 - Epithelial Na⁺ channel (ENaC)
 - Intercalated cells (~30%): Some K⁺ reabsorption from the tubular fluid
- Early portion of the DCT secrete K⁺ at a low rate.
- Late portions of the classic distal tubule (i.e., CT, CCT) have a high rate of K⁺ secretion.
- Principal Cell:
 - Mainly responsible for K⁺ secretion
 - The three key elements:
 - Na⁺-K⁺ ATPase for K⁺ (basolateral membrane).
 - High & variable apical K⁺ permeability (ROMK channel)
 - EC driving force for K⁺ exit across the apical membrane.
 - K⁺ can also move to lumen via an apical K⁺/Cl⁻ cotransporter (KCC).
- Net effect: Active movement of basolateral K⁺ → cell → passive movement to lumen (urine).
- Changes in any of the three elements can affect the secretion of K⁺.

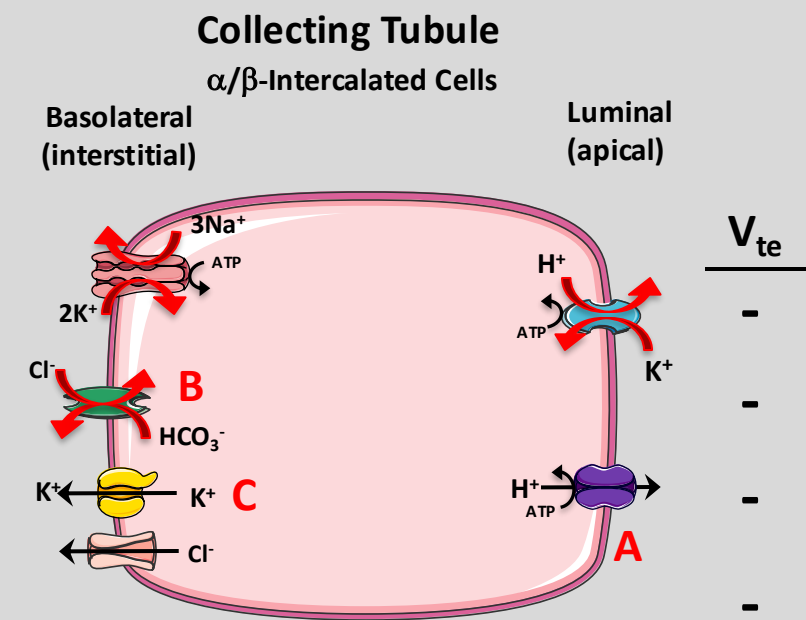
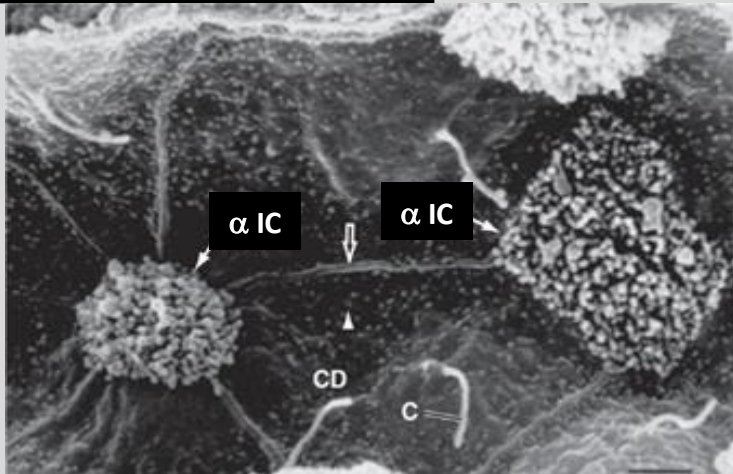
Collecting Tubule Principal Cells



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K⁺ Reabsorption in the Collecting Tubule

- Intercalated cells:
 - Type A (α) Intercalated Cells: 1^o role in H⁺ secretion (**A**)
 - Type B (β) Intercalated Cells: 1^o role in HCO₃⁻ reabsorption (**B**)
 - Morphologically distinct from the Principal cells
 - ICT, CCT, and MCD reabsorb K⁺ in response to K⁺ depletion (transcellular reabsorption).
- Reabsorption occurs in 2 steps:
 - Active step mediated by an apical H⁺/K⁺ ATPase (**A**)
 - Passive step mediated by a basolateral K⁺ channel (allows leakage) (**C**)
- K⁺ depletion produces an increase in H⁺/K⁺ ATPase density & activity.
- This will enhance K⁺ reabsorption.
- Increase in H⁺/K⁺ ATPase activity accelerates H⁺ secretion
- This contributes to the development of HYPOKALEMIC ALKALOSIS.



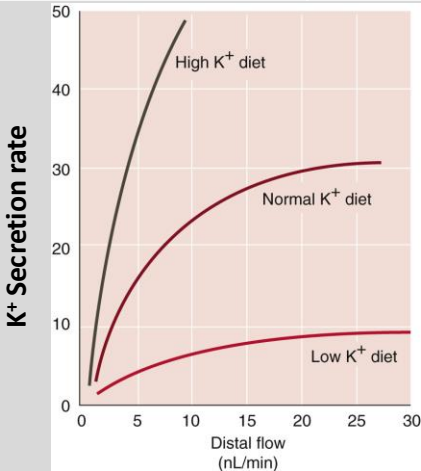
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Factors Modulating Secretion by the Distal K⁺-Secretory System

- K⁺ reabsorption along the MCD is both passive and active.
- Capacity for K⁺ secretion diminishes from cortex to MCD.
- K⁺ loss from the MCD lumen can occur by passive movement via the paracellular pathway driven by [K⁺] gradient.
- Luminal [K⁺] of the MCD is high for:
 - Upstream segments (distal K⁺-secretory system) secrete K⁺.
 - Fluid reabsorption of fluid in the medullary collecting tubules in the presence of high arginine vasopressin (AVP) levels increase luminal K⁺.
- In addition to passive K⁺ reabsorption, apical H⁺/K⁺ ATPase mediate K⁺ reabsorption, especially during low K⁺ intake.
- Fluid flow:
 - Very potent stimulus of K⁺ secretion along the ICT and CCT.
 - Similar relationship holds between urine flow and K⁺ excretion (kaliuresis).
 - Increased urine flow (e.g., EC volume expansion, osmotic diuresis, or administration of diuretic agents) → Enhanced kaliuresis.
- Increased luminal flow also increases Na⁺ in tubule cells
- This ↑luminal [Na⁺] and enhances Na⁺ uptake by principal cells
- This stimulates H⁺/K⁺ ATPase activity → ↑K⁺ secretion.
- ↑urinary flow is associated with increased natriuresis

LUMINAL FACTORS	
STIMULATORS	INHIBITORS
↑ Flow Rate	↑ [K ⁺]
↑ [Na ⁺]	↑ [Cl ⁻]
↓ [Cl ⁻]	↑ [Ca ⁺⁺]
Negative luminal voltage	Barium
Diuretics acting upstream	Amiloride

PERITUBULAR FACTORS	
STIMULATORS	INHIBITORS
↑ K ⁺ Intake	↓ pH
↑ [K ⁺]	Epinephrine
↑ pH	
Mineralocorticoids	
Arginine vasopressin	

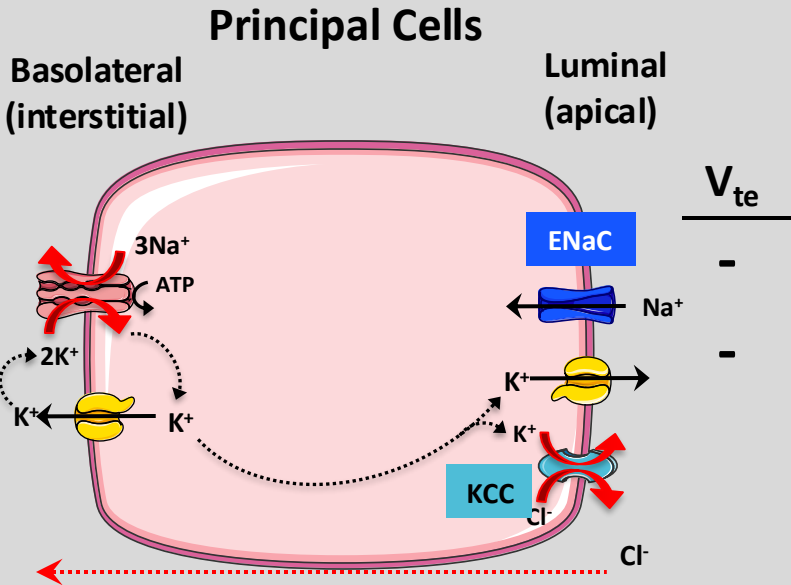


Transepithelial Potential & K⁺ Secretion

- K⁺ secretion in the ICT and CCT occurs by diffusion of K⁺ from the principal cell to the lumen.
- Process depends on the apical electrochemical K⁺ gradient.
- Increases in luminal [Na⁺] enhance apical Na⁺ entry through epithelial Na⁺ channels (ENaCs).
- This depolarizes the apical membrane, favoring the exit of K⁺ from cell to lumen.
- Conversely, a fall in luminal [Na⁺] will hyperpolarize the apical membrane
- This will inhibit K⁺ secretion.
- K⁺-sparing diuretics (e.g., amiloride) block apical ENaCs
- Same effect on K⁺ secretion as decreasing luminal [Na⁺].
- This hyperpolarizes the apical membrane & reduces E.C. gradient for K⁺ secretion.
- Inhibition of apical Na⁺ will lower ↓[Na⁺] and reduce Na⁺-K⁺ ATPase activity.
- Lowering luminal [Cl⁻] & replacing it with poorly-reabsorbed anion (e.g., SO₄⁻², HCO₃⁻) promotes urinary K⁺
- This is independent of changes in the lumen-negative V_{te}.
- Lowering luminal [Cl⁻] increases the cell → lumen Cl⁻ gradient
- This stimulates K⁺/Cl⁻ cotransporter (KCC) and K⁺ secretion.

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STIMULATORS	INHIBITORS
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↑ [Na ⁺]	↑ [Cl ⁻]
↓ [Cl ⁻]	↑ [Ca ⁺⁺]
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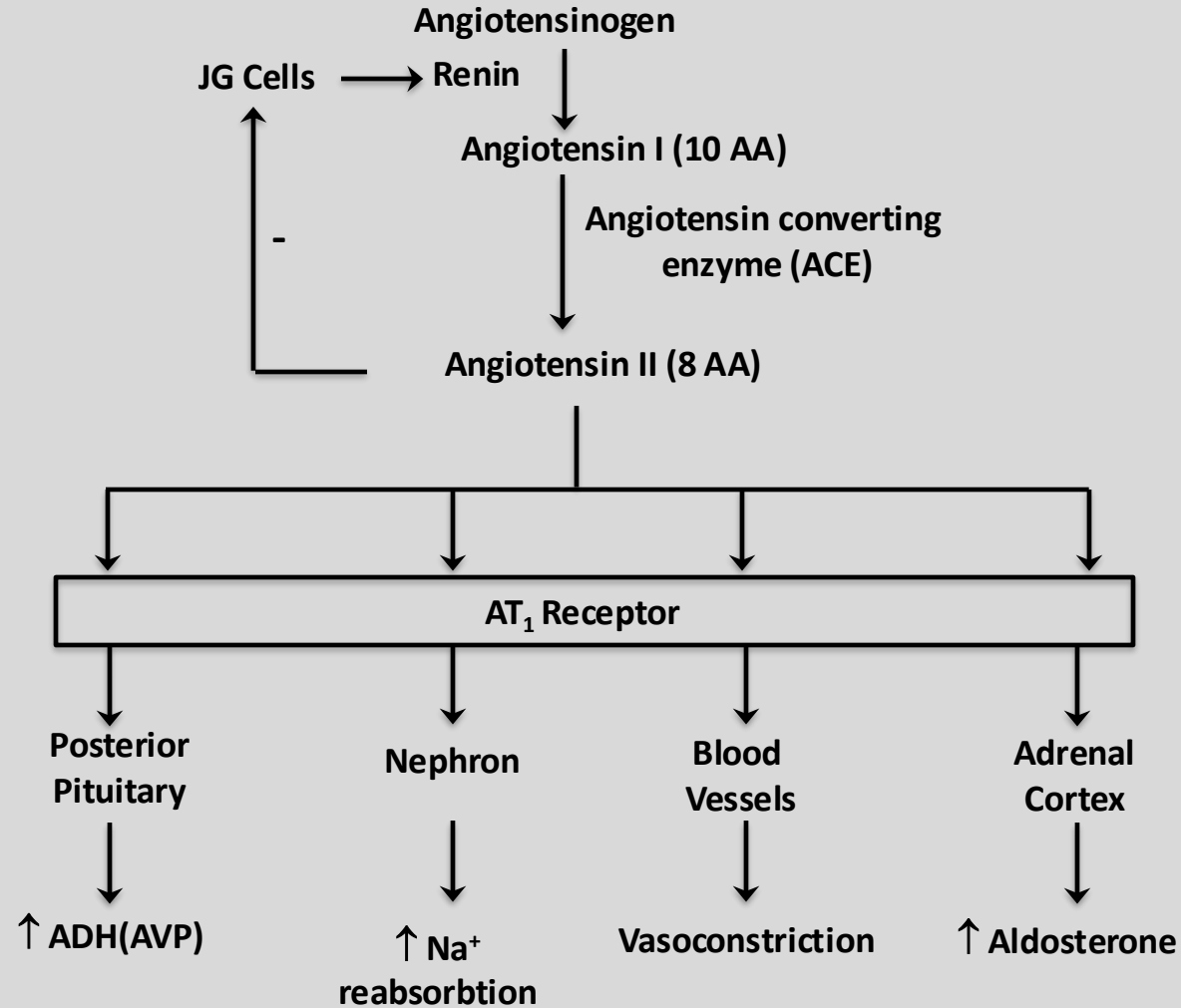
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STIMULATORS	INHIBITORS
↑ K ⁺ Intake	↓pH
↑ [K ⁺]	Epinephrine
↑ pH	
Mineralocorticoids	
Arginine vasopressin	



Renin Angiotensin-Aldosterone System

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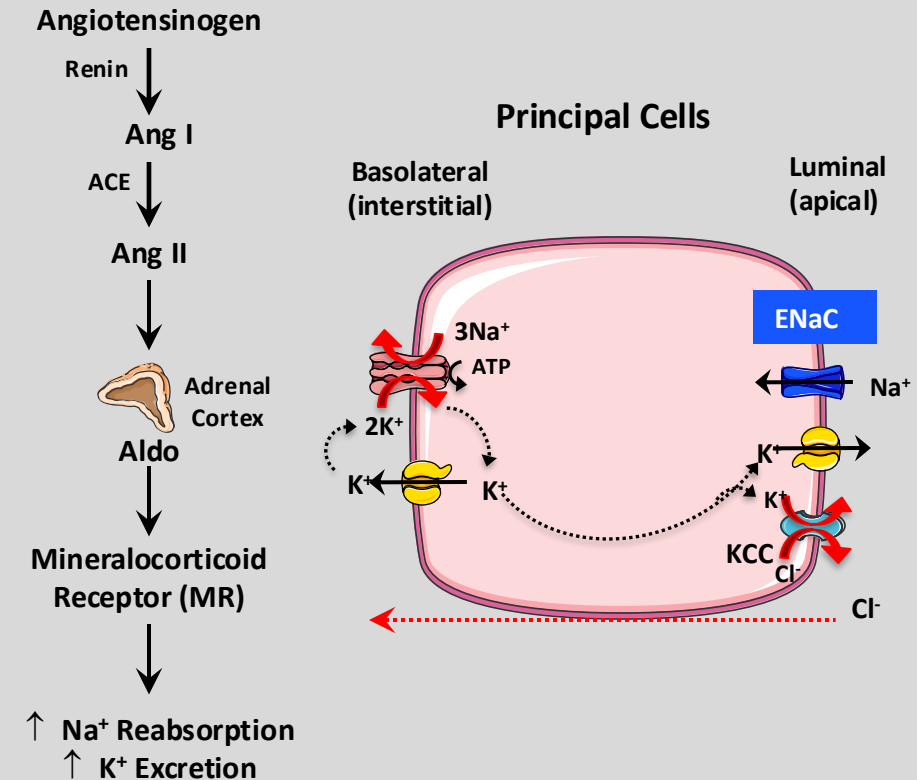
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Corticosteroids & K⁺ Secretion

- The adrenal cortex is the major site of steroid hormone synthesis (glucocorticoids (GC), mineralocorticoids (MC), adrenal androgens (17-ketosteroids), estrogens, and progestogens).
- Aldosterone (Aldo) is the MAJOR mineralocorticoid receptor (MR) produced.
- Secretion of Aldo from the zona glomerulosa of the adrenal cortex is regulated by Ang II, K⁺, and (to a lesser extent) ACTH.
- The renin-angiotensin-aldosterone system (RAAS) is the major regulator of renal sodium excretion and blood pressure.
- Aldo induces K⁺ secretion in the ICT and CCT via the transcription of ENaC and ROMK.
- Aldo also increases K⁺ uptake by stimulating Na⁺-K⁺ ATPase activity & #.
- Extent to which Aldo increases K⁺ excretion depends on Na⁺ excretion (e.g. low Na⁺ intake and excretion)
- Under physiological conditions, GC enhance K⁺ excretion.
- Natural & synthetic GC (corticosterone & dexamethasone, respectively) produce this by increasing flow along the ICT and CCT.
- GCs also increase GFR, increasing amounts of H₂O & Na⁺ in these segments.
- Unphysiologically high [GC] stimulate K⁺ by binding & activating MR's.

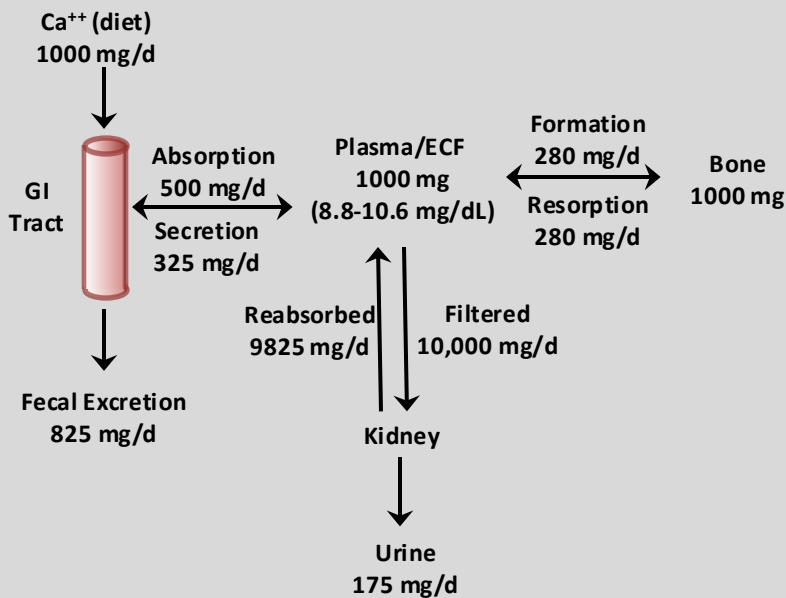
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STIMULATORS	INHIBITORS
↑ K ⁺ Intake	↓ pH
↑ [K ⁺]	Epinephrine
↑ pH	
Mineralocorticoids	
Arginine vasopressin	



Renal Regulation of Ca⁺⁺

- Ca⁺⁺ plays a critical role in secretion, contraction, conduction, exocytosis, and the activation and inactivation of enzymatic pathways.
- [Ca⁺⁺]_{pl} is strictly regulated
- Protein-bound Ca⁺⁺: Bound to plasma proteins (mostly albumin) & are nonfilterable.
- The filterable portion: Diffusible Ca⁺⁺ complexes (bound to CO₃⁻², citrate, PO₄³⁻ & SO₄³⁻) & free, ionized Ca⁺⁺.
- Free, ionized Ca⁺⁺ is tightly regulated.
- Distribution of various forms of total calcium is not fixed.
- H⁺ competes with Ca⁺⁺ for binding to anionic sites on proteins or small molecules
- Acidosis: Increases plasma [Ca⁺⁺]_{pl}
- Alkalosis: Acutely reduces [Ca⁺⁺]_{pl}
- Acute alkalosis can cause signs and symptoms mimicking hypocalcemia.
- Ca⁺⁺ binding to plasma proteins depends on the concentrations of the proteins.
- 99% is stored in bone
- Chemically linked with other components of hydroxyapatite [Ca₅(PO₄)₃ (OH)]
- Hypercalcemia: Multiple effects (effects on excitable cell function)
- Hypercalcemia: CNS depression, muscle weakness, and GI tract immobility.

Ca⁺⁺ Distribution & Balance



	mg/dL	mM	% Total
Ionized Ca ⁺⁺	4.7	1.2	~ 45
Diffusible Ca ⁺⁺ complexes	1.4	0.3	~15
Protein-bound Ca ⁺⁺	3.9	1.0	~40
Total Ca ⁺⁺	10.0	2.5	100

Ca⁺⁺ Transport in the Nephron

- 99% of the filtered Ca⁺⁺ is reabsorbed.
- Occurs in the PCT, tALH & DCT
- PCT & TALH: Reabsorption is tightly coupled to Na⁺ reabsorption
- DCT: Reabsorption is dissociated from Na⁺.

Proximal Tubule:

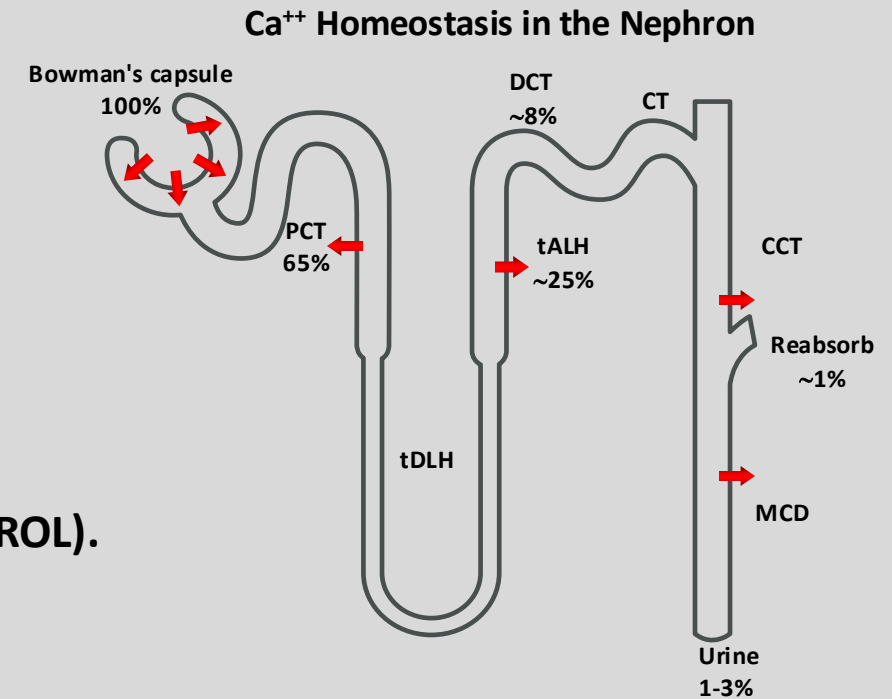
- Reabsorbs ~ 65% of the filtered calcium (NOT SUBJECT TO HORMONAL CONTROL).
- All PCT Ca⁺⁺ reabsorption occurs via the paracellular routes.

PCT:

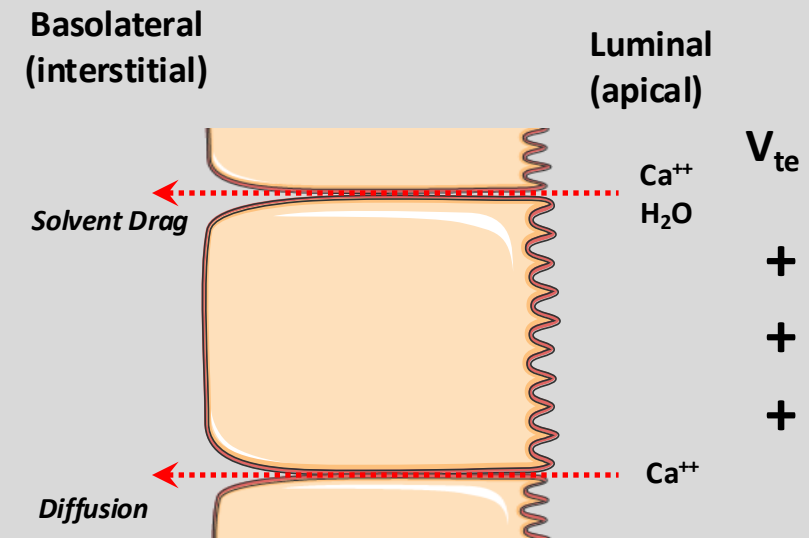
- High Ca⁺⁺ permeability
- Transport is sensitive to changes in V_{te}
- Lumen-negative $V_{te} \rightarrow$ Ca⁺⁺ secretion
- Lumen-positive $V_{te} \rightarrow$ Ca⁺⁺ reabsorption.
- Paracellular reabsorption: Dependent on luminal [Ca⁺⁺].

H₂O reabsorption $\rightarrow \uparrow [Ca^{++}]_L \rightarrow$ favorable $V_{te} \rightarrow$ Ca⁺⁺ reabsorption.

- The greater the fluid reabsorption along the PT, the greater the reabsorption
- $\uparrow$ NaCl diet $\rightarrow \uparrow$ ECV $\rightarrow \downarrow$ Na⁺ & H₂O reabsorption $\rightarrow \downarrow$ Ca⁺⁺ reabsorption $\rightarrow$ calciuria



Proximal Tubule



Ca⁺⁺ Transport in the Nephron

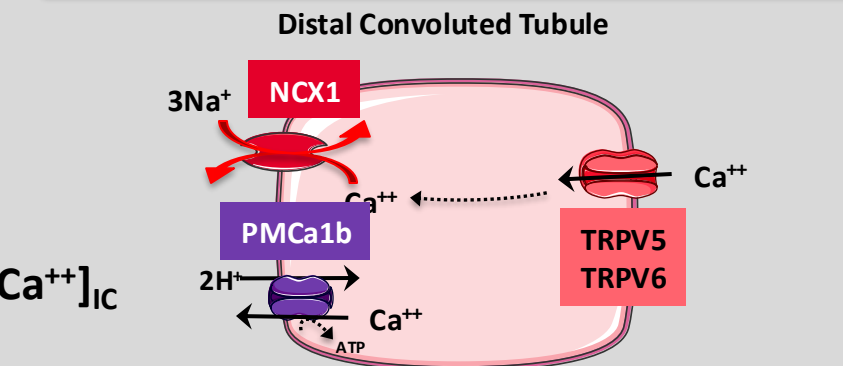
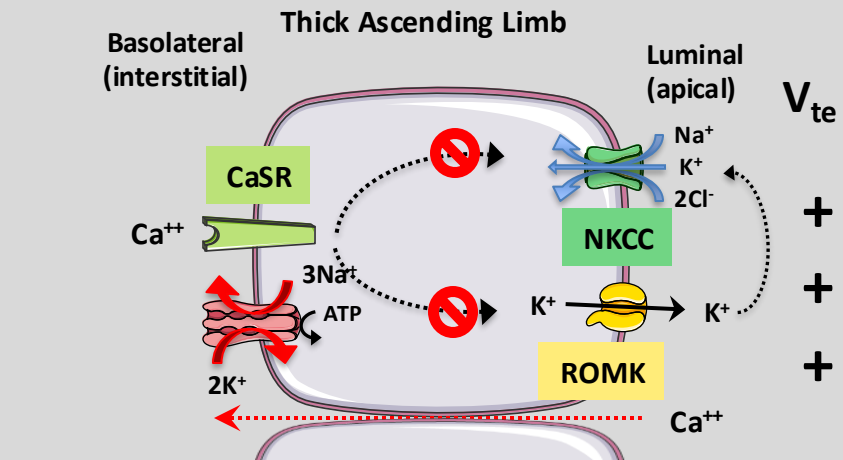
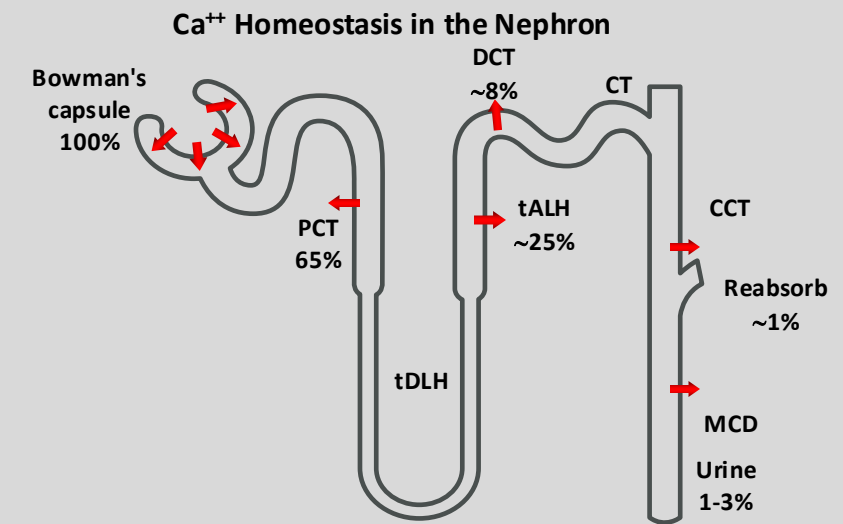
tALH:

- Reabsorbs ~ 25% of the FL of Ca⁺⁺.
- Most of the Ca⁺⁺ reabsorption occurs passively via a paracellular route
- Driven by the lumen-positive V_{te}.
- [Ca⁺⁺] can also bind to a basolateral Ca⁺⁺-sensing receptor (CaSR)
- Results in inhibition of ROMK & NKCC → ↓ lumen-positive V_{te}
- ↓ Ca⁺⁺ reabsorption & and ↑ Ca⁺⁺ excretion.

DCT:

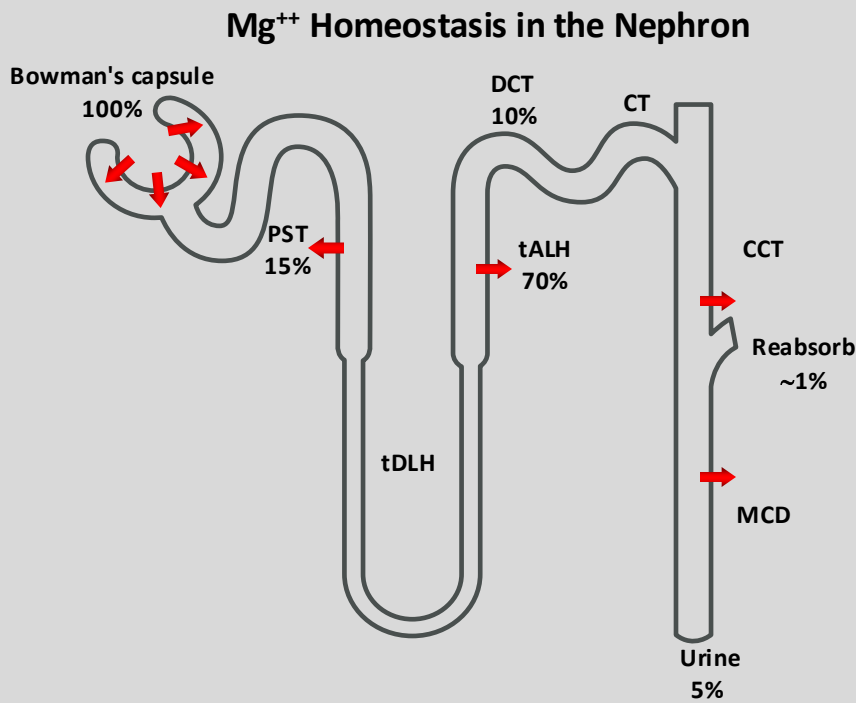
Reabsorbs ~ 8% of the FL of Ca⁺⁺

- Major regulatory site despite small amount of Ca⁺⁺.
- Reabsorption is predominantly via active, transcellular route.
- MP of DCT ~ -70 mV (steep electrochemical gradient)
- Apical Ca⁺⁺ is passive, mediated by the epithelial Ca⁺⁺ channels (TRPV5 & TRPV6).
- DCT cells must extrude across the basolateral membrane all Ca⁺⁺ entering cells
- Both 1° (Ca⁺⁺-ATPase, PMCa1b) & 2° active transporters participate.
- The Na⁺-Ca⁺⁺ exchanger (NCX1) extrudes Ca⁺⁺ (a bi-directional exchanger)
- ↓ [Ca⁺⁺]_{pl} → low NCX1 activity → Ca⁺⁺-ATPase is the major pathway.
- NCX1 begins to make a significant contribution when ↑ [Ca⁺⁺]_{pl}
- Normal [Ca⁺⁺]_{pl} & EC gradient falls (e.g., ↑ [Na⁺]_{ic}) → NCX1 reverse direction → ↑ [Ca⁺⁺]_{ic}



Renal Regulation of Mg^{++}

- Body maintains the total $[Mg^{++}]_{Pl}$ w/i narrow limits (0.8-1.0 mM)
- ~99% of the total body stores of Mg^{++} reside within:
 - Bone (~54%)
 - I.C compartment (~45%)
- Not chemically linked with other components of hydroxyapatite.
- Filterable Portion = Ionized Mg^{++} + Diffusible Mg^{++} complexes (PO_3^{-2} , citrate, oxalate)
- Disturbances of Mg^{++} metabolism usually involve abnormal losses
- Occur most frequently during GI malabsorption, diarrhea, renal disease & diuretic use.
- Clinical manifestations:
 - Neurological disturbances (e.g., tetany (especially when associated with hypocalcemia), cardiac arrhythmias, $\uparrow$ PVR.
- Increased Mg^{++} intake:
 - $\downarrow$ BP
- Severe hypermagnesemia, following excessive intake or renal failure, results in a toxic syndrome involving nausea, hyporeflexia, respiratory insufficiency, and cardiac arrest.



	mg/dL	mM	% Total
Ionized Mg^{++}	1.3	0.56	62
Diffusible Mg^{++} complexes	0.1	0.06	7
Protein-bound magnesium	0.6	0.28	31
Total magnesium	2.0	0.90	100

Mg⁺⁺ Transport in the Nephron

- In contrast to the predominantly PT reabsorption pattern of ultrafiltrate, majority of Mg⁺⁺ reabsorption occurs along tALH.

PT:

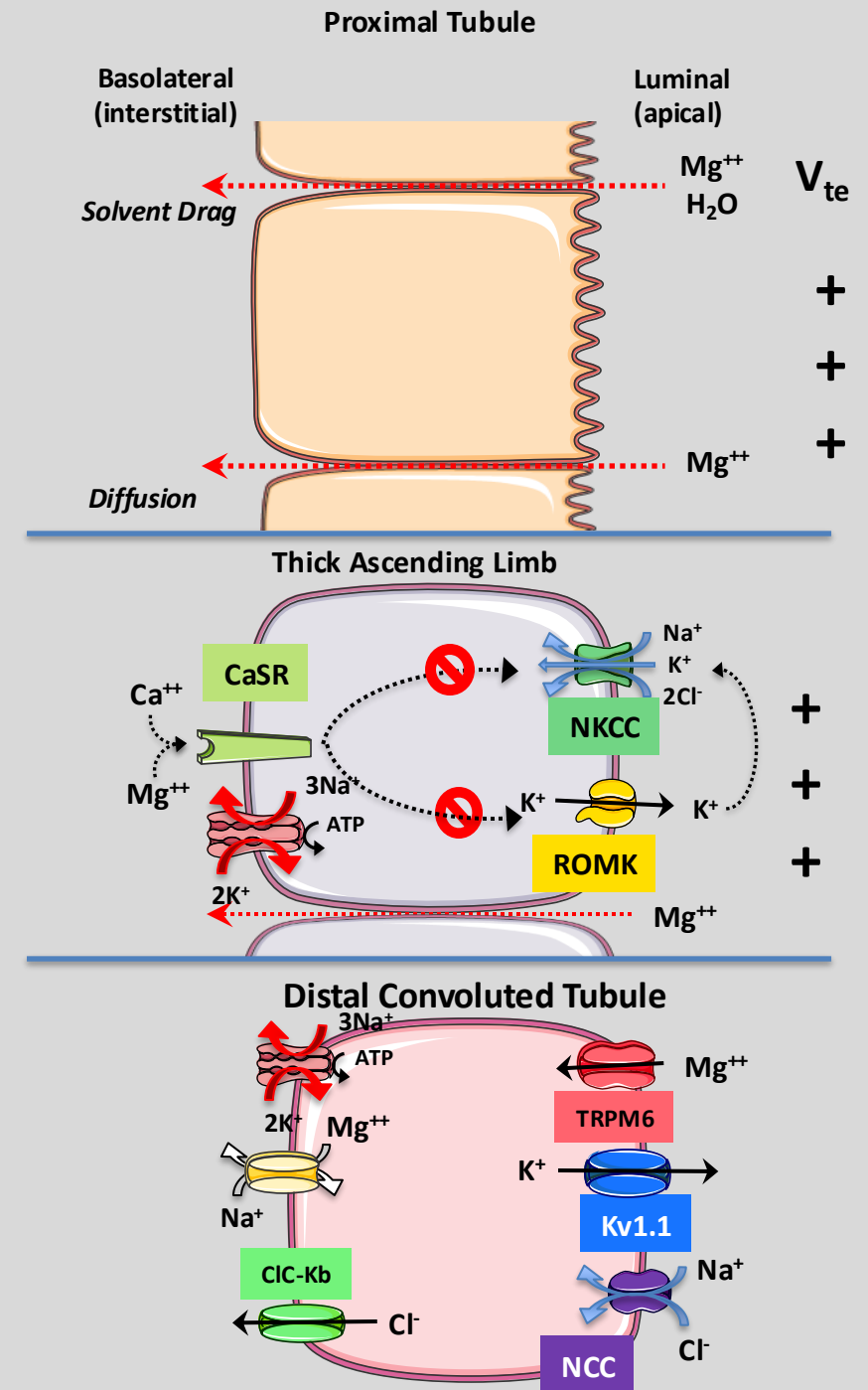
- Reabsorbs ~15% of the FL of Mg⁺⁺.
- H₂O reabsorption in the PT causes [Mg⁺⁺]_L to 2x vs [Mg⁺⁺]_{BC} Bowman's space
- Favors [Mg⁺⁺] EC gradient for paracellular reabsorption.
- Solvent drag contributes to paracellular reabsorption.

tALH:

- Reabsorbs ~70% of the FL of Mg⁺⁺.
- Lumen-positive V_{te} is driving force for paracellular reabsorption.
- Tight-junction proteins claudin 16 & 19 (CLDN16 & CLDN19 genes) account for the high paracellular permeability.
- CLDN16 or CLDN19 gene mutations cause an autosomal recessive disorder of renal Mg⁺⁺ wasting.

DCT:

- Reabsorption along the DCT is predominantly transcellular
- ~10% of the FL of Mg⁺⁺.
- K⁺ efflux neutralizes the entry of positive charge via Mg⁺⁺ uptake.
- Reabsorption is also stimulated by activation of epidermal growth factor receptor (EGFR).



Factors Affecting Mg^{++} Reabsorption

Nephron Segment	Increase	Decrease
PT	Volume Contraction	Volume Expansion
tALH	PTH, Calcitonin, Glucagon, AVP/ADH, Hypomagnesemia, Metabolic alkalosis	Loop Diuretic, Osmotic Diuretics, Hypermagnesemia, Hypercalcemia
DCT	PTH, Calcitonin, Glucagon, Estrogens, Hypomagnesemia, K-sparing diuretics	Hypermagnesemia, Hypercalcemia, Metabolic Acidosis, Hypokalemia, Hypophosphatemia, Thiazide Diuretics

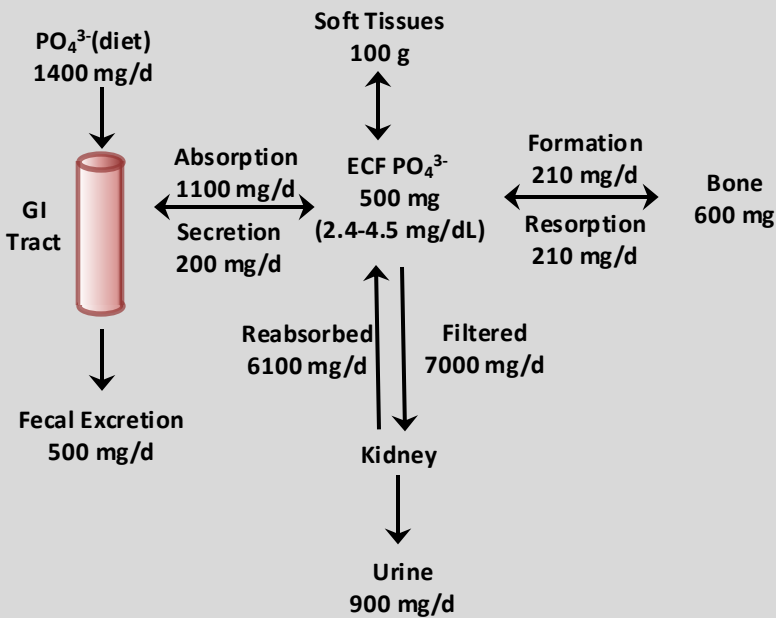
Regulation of Mg^{++} Reabsorption

- Mg^{++} reabsorption is stimulated by the epidermal growth factor receptor (EGFR).
- Renal Mg^{++} -wasting syndromes:
 - Mutations in TRPM6, Kv1.1, and EGF.
- Effective Circulating Volume (ECV):
 - $\downarrow ECV \rightarrow \uparrow Na^+$ reabsorption & $\uparrow EC$ force $\rightarrow \uparrow Mg^{++}$ reabsorption.
 - $\uparrow ECV \rightarrow \downarrow Na^+$ reabsorption & $\downarrow EC \rightarrow \downarrow Mg^{++}$ reabsorption.
- Acid-Base Balance: Identical to Ca^{++}
 - Alkalosis: $\uparrow Mg^{++}$ reabsorption ($\downarrow Mg^{++}$ excretion)
 - Acidosis: $\downarrow Mg^{++}$ reabsorption ($\uparrow Mg^{++}$ excretion)
- Hypomagnesemia:
 - Reduction of fractional excretion of Mg^{++}
- Hypermagnesemia/Hypercalcemia:
 - Kidneys discriminate poorly between $\uparrow [Mg^{++}]_{PI} / \uparrow [Ca^{++}]_{PI}$.
 - Each will $\downarrow$ reabsorption of both Mg^{++} & Ca^{++} due to binding to CaSR.
- Hormones:
 - PTH: Most important hormone for Mg^{++} regulation
 - AVP, glucagon & calcitonin: $\uparrow$ reabsorption (TAL)
- Diuretics:
 - In general, diuretics $\downarrow Mg^{++}$ reabsorption & $\uparrow$ excretion.

Renal Regulation of PO_4^{3-}

- PO_4^{3-} is no less important than Ca^{++} & plays a critical role in many cellular processes.
- Part of ATP → Critical role in cellular energy metabolism.
- $[\text{PO}_4^{3-}]_{\text{PI}}$ is not strictly regulated.
- Levels fluctuate throughout the day, particularly after meals.
- Ca^{++} & PO_4^{3-} homeostasis are:
 - Principal components of hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$ (major portion of the mineral phase of bone)
 - Regulated by parathyroid hormone (PTH), 1,25-dihydroxyvitamin D (calcitriol) & calcitonin.
- Total $[\text{HPO}_4^{-2}]_{\text{PI}}$ & $[\text{H}_2\text{PO}_4^{-}]_{\text{PI}}$ in adults: 0.8-1.5 mM.
- ~ 50% higher in children.
- Ionized phosphates & Diffusible phosphate complexes (complexed to Na^+ , Ca^{+2} , or Mg^{+2} are filtered by the kidneys
- Soft tissues phosphates: Mainly organic phosphates (e.g., PL, phosphoproteins, nucleic acids & nucleotides
- The kidneys filter ~14 times the total E.C pool of phosphate per day (~7000 mg/day) and reabsorb (~6100 mg/day)
- Net renal excretion of phosphorus is the same as the net absorption by the GI tract.

PO_4^{3-} Distribution & Balance

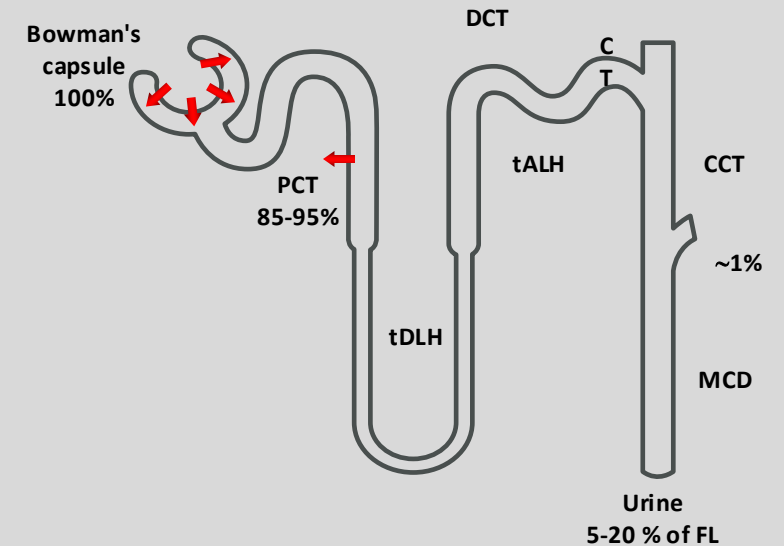


	mg/dL	mM	% Total
Ionized HPO_4^{-2} & $\text{H}_2\text{PO}_4^{-}$	2.1	0.7	50
Diffusible phosphate complexes	1.5	0.5	40
Protein-bound phosphates	0.6	0.2	10
Total phosphate	4.2	1.4	100

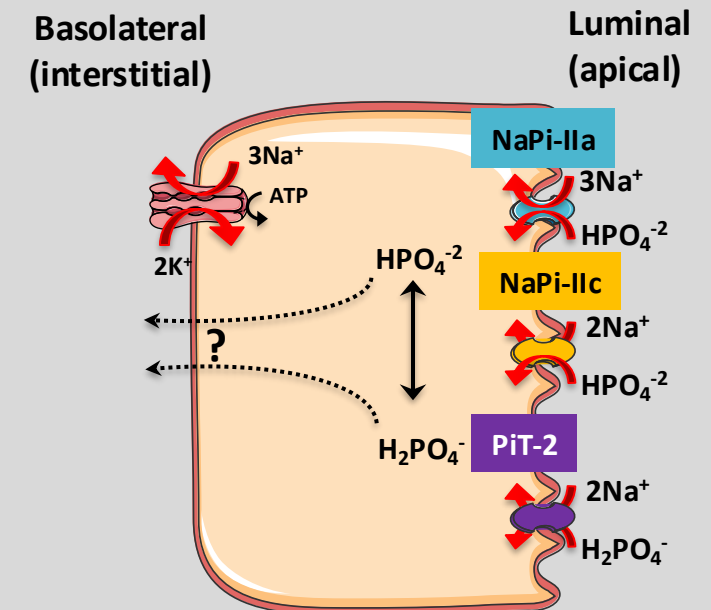
HPO_4^{-2} & $\text{H}_2\text{PO}_4^{-}$ Transport in the Nephron

- The proximal tubule reabsorbs 80% to 95% of $\text{HPO}_4^{-2}/\text{H}_2\text{PO}_4^{-}$ FL via transcellular route
- Distal nephron segments reabsorb a negligible fraction of the FL.
- Three separate Na^+ /phosphate cotransporters contribute: NaPi-IIa, NaPi-IIc
- Renal phosphate transport is regulated by:
 - Hormones
 - Parathyroid hormone (PTH)
 - Dopamine
 - 1,25-dihydroxy-vitamin D
 - Phosphatonins (e.g., FGF23): Circulating factors that cause renal phosphate wasting in disease (e.g., oncogenic osteomalacia, autosomal dominant & X-linked hypophosphatemic rickets).
 - Glucocorticoids
 - Acid-base balance

HPO_4^{-2} & $\text{H}_2\text{PO}_4^{-}$ Homeostasis in the Nephron



Proximal Tubules



NEW YORK INSTITUTE OF TECHNOLOGY

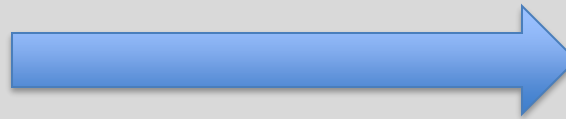
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Hot
Yoga



Excessive
sweating

K⁺ loss in sweat



Dizziness
Funny
Heartbeats?



Nutrition Facts (Refrigerate)	
Serving Size	10 fl oz
Amount Per Serving	
Calories	80
	% Daily Value*
Total Fat 0g	0%
Saturated Fat 0g	0%
Trans Fat 0g	
Cholesterol 0g	0%
Sodium 50mg	2%
Total Carbohydrate 19g	7%
Dietary Fiber 0g	0%
Sugar 17g	0%
Added Sugar 0g	0%
Protein 0g	0%
Vitamin D 0mg	0%
Potassium 677mg	15%
Calcium 5mg	0%
Iron 0mg	0%

* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.



BANANAS

"The Potassium Power Fruit"

HEALTHY BLOOD
STRONG IMMUNE
SYSTEM

REDUCE
STRESS

HELP
WITH
DIGES-
TION



VITAMIN B6.....25.2%
MANGANESE.....16%
VITAMIN C.....13.6%
FIBER.....12.2%
POTASSIUM.....12%
BIOTIN.....10.2%

DID YOU KNOW? FREEZING A BANANA CAN KEEP IT FRESH FOR UP TO 2 MONTHS!



Lecture Feedback Form

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

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